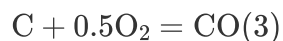
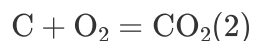
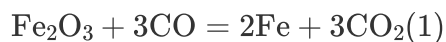


## Question

Known that in the process of smelting a certain metal in a blast furnace, the reactions involved are as follows:



The  $\Delta_f H_m^\ominus$  of each substance involved above are as follows:

$$\Delta_f H_m^\ominus (\text{Fe}_2\text{O}_3) = -824.25 \text{ kJ} \cdot \text{mol}^{-1}$$

$$\Delta_f H_m^\ominus (\text{CO}_2) = -393.52 \text{ kJ} \cdot \text{mol}^{-1}$$

$$\Delta_f H_m^\ominus (\text{CO}) = -110.53 \text{ kJ} \cdot \text{mol}^{-1}$$

It is now known that at the working temperature of the blast furnace, in the tail gas of the reaction between CO and  $\text{Fe}_2\text{O}_3$ ,  $p(\text{CO})/p(\text{CO}_2) = 0.40$ . If every 1 mol Fe is produced, 400.00 kJ of heat will be dissipated from the furnace wall to the air. The system has no other heat sources or material exchange, and the system is a constant pressure system. If the furnace temperature is kept constant, please calculate the  $p(\text{CO})/p(\text{CO}_2)$  in the products of coke combustion, the correct option is:

**A.** All other options are incorrect

**B.** 3.01

**C.** 6.31

**D.** 1.22

**E.** 1.00

**F.** 3.36

**G.** 5.50

**H.** 3.57

**I.** 5.88

## Answer

Correct Answer: C

## Detailed Explanation

$$\Delta_r H_m^\circ(1) = 3 \times \Delta_f H_m^\circ(\text{CO}_2) - 3 \times \Delta_f H_m^\circ(\text{CO}) - \Delta_f H_m^\circ(\text{Fe}_2\text{O}_3) = -24.72 \text{ kJ} \cdot \text{mol}^{-1}$$

### CHECKPOINT

1 PTS

$$\Delta_r H_m^\circ(1) = -24.72 \text{ kJ} \cdot \text{mol}^{-1}$$

Let  $p(\text{CO})/p(\text{CO}_2) = x$  .

From this, it can be seen that whenever there is 1 mol  $\text{CO}_2$  in the combustion products,  $x$  mol  $\text{CO}$  will be generated. Therefore, 1 mol of reaction (2)

and  $x$  mol of reaction (3) will occur.

In addition, assuming that  $y$  mol of reaction (1) occurs, then in the exhaust gas,  $p(\text{CO})/p(\text{CO}_2) = (x - 3y)/(1 + 3y) = 0.40$  and heat is dissipated

$800.00y$  kJ .

### CHECKPOINT

1 PTS

In the exhaust gas,  $p(\text{CO})/p(\text{CO}_2) = (x - 3y)/(1 + 3y) = 0.40$  and heat is dissipated  $800.00y$  kJ

Since the furnace temperature remains unchanged, the sum of the total enthalpy change of the three reactions is 0. Among them, the enthalpy changes of reaction (2) and reaction (3) are the enthalpy of formation of  $\text{CO}_2$  and  $\text{CO}$ :

$$-393.52 - 110.53x - 24.72y + 800.00y = 0$$

CHECKPOINT

1 PTS

Thermal equilibrium:  $-393.52 - 110.53x - 24.72y + 800.00y = 0$

The above two equations can be solved simultaneously to obtain:

$$x = 6.3$$

CHECKPOINT

1 PTS

$$p(\text{CO})/p(\text{CO}_2) = 6.3$$